

Predicting Profitability in Logistics Operations Using ML

Maryi Tatiana Palacios Giraldo
Data Science
Regis University
Regis University, Denver, CO, USA
mpalaciosgiraldo@regis.edu

Abstract

Profitability is a key performance indicator in logistics operations, directly influencing operational efficiency, resource utilization, and business sustainability. At Gallagher Transport International, thousands of shipments are processed each year across multiple transportation modes and international routes. Although large amounts of operational data are generated during shipment execution, profitability is typically evaluated only after shipment completion. Consequently, operational decisions rely primarily on historical experience rather than predictive analytics, limiting the organization's ability to proactively identify shipments that may result in lower profitability.

This practicum project develops and evaluates a machine learning framework to predict shipment profitability using historical operational data from Gallagher Transport International. The dataset includes operational information such as transportation mode, shipment characteristics, origin and destination locations, processing timelines, financial outcomes, and shipment status. Data preprocessing involved cleaning, feature engineering, and class imbalance handling using the Synthetic Minority Over-sampling Technique (SMOTE), while sensitive customer information was excluded to ensure ethical data handling and privacy protection.

The project involves data cleaning, exploratory data analysis, feature engineering, and the development of multiple machine learning classification models to predict shipment profitability. Logistic Regression, Random Forest, and XGBoost were evaluated and compared against a Baseline classifier, while SMOTE was applied to address class imbalance and GridSearchCV was used to optimize the Random Forest model. The results identified Random Forest as the best-performing algorithm, achieving approximately 92

This project demonstrates the practical application of machine learning within a real-world logistics environment by integrating multiple stages of the data science lifecycle, including data preparation, exploratory analysis, predictive modeling, model evaluation, and business interpretation. The proposed framework provides a practical decision-support tool that enables organizations to estimate shipment profitability before execution, supporting more informed operational planning and resource allocation.

I. INTRODUCTION/BACKGROUND

This practicum project originated from operational challenges identified during daily work activities at Gallagher Transport International, an international customs brokerage and freight logistics company specializing in household goods and commercial shipments. The company manages thousands of shipments across multiple transportation modes and international trade routes, requiring continuous coordination among customs brokers, logistics personnel, and clients. These operations generate large volumes of operational and financial data, creating valuable opportunities for applying data-driven approaches to improve business performance.

During regular operational activities, it became evident that shipment profitability varies considerably across shipments, even when they share similar transportation modes or operational characteristics. Despite the availability of extensive historical shipment data, the organization currently lacks a centralized analytical framework capable of identifying the operational factors associated with profitability or predicting profitable shipment outcomes before shipment completion.

At present, shipment profitability is typically assessed only after shipment completion, limiting the organization's ability to make proactive operational decisions. Without a predictive approach, managers cannot identify shipments with a higher risk of low profitability before execution or understand how operational factors influence financial outcomes. Additionally, the absence of structured predictive analysis limits the organization's ability to leverage historical shipment data to improve planning, resource allocation, and overall operational efficiency.

The purpose of this practicum project is to develop and evaluate a machine learning framework capable of predicting shipment profitability using historical operational data from international logistics operations. The project integrates multiple stages of the data science lifecycle, including data preparation, exploratory data analysis, feature engineering, class imbalance handling, predictive modeling, hyperparameter optimization, model evaluation, and business interpretation. By comparing multiple machine learning algorithms, the study aims to identify the most effective approach for supporting profitability prediction before shipment completion.

This project extends beyond descriptive analysis by applying machine learning to support operational decision-making in logistics. Specifically, it provides a predictive framework for estimating shipment profitability before shipment completion, enabling organizations to identify potentially low-profit shipments and better understand the operational factors that influence financial performance. By transforming historical shipment data into actionable business insights, the proposed approach supports more informed planning, resource allocation, and profitability management within logistics operations.

II. PROBLEM STATEMENT

At Gallagher Transport International, shipment operations are managed daily across household goods and commercial logistics services. Although the company generates extensive operational and financial data throughout the shipment lifecycle, profitability is typically assessed only after shipment completion. As a result, operational decisions are often made without predictive insight into which shipments are likely to be profitable or unprofitable, limiting the organization's ability to proactively optimize operational and financial performance.

Although the organization processes thousands of shipments each year, historical operational data has not been systematically leveraged to identify the factors that influence shipment profitability or to support predictive decision-making. Consequently, managers have limited visibility into which operational characteristics contribute to profitable or unprofitable shipments before execution. This limitation reduces the organization's ability to proactively optimize resource allocation, improve operational efficiency, and support profitability-driven business decisions.

The central problem addressed in this practicum project is the absence of a data-driven framework capable of predicting shipment profitability using historical operational data. This study seeks to determine whether machine learning algorithms can accurately classify shipments as profitable or non-profitable before shipment completion based on operational characteristics available during the planning stage. Additionally, the project aims to identify the operational factors that have the greatest influence on shipment profitability, providing actionable insights to support data-driven decision-making in logistics operations.

The absence of a data-driven profitability prediction framework limits the organization's ability to proactively identify shipments with a higher risk of low profitability before execution. As a result, operational decisions related to shipment planning and resource allocation are made without predictive insight into potential financial outcomes. Implementing a machine learning framework can support more informed decision-making by enabling early profitability estimation, improving operational efficiency, and enhancing overall business performance.

Research Questions:

- 1) Can shipment profitability be accurately predicted using historical operational data from international logistics operations?
- 2) Which operational characteristics have the greatest influence on shipment profitability?
- 3) How do different machine learning algorithms compare in predicting shipment profitability?
- 4) How can profitability prediction support operational planning and data-driven decision-making in logistics operations?

Hypotheses

- 1) **H1:** Historical operational shipment data can be used to accurately predict shipment profitability.
- 2) **H2:** Tree-based machine learning algorithms, such as Random Forest and XGBoost, will outperform Logistic Regression in predicting shipment profitability.
- 3) **H3:** Balancing the training dataset using the Synthetic Minority Over-sampling Technique (SMOTE) will improve the model's ability to identify the minority class without substantially reducing overall predictive performance.
- 4) **H4:** Temporal, operational, and shipment-related characteristics significantly influence shipment profitability.

III. RELATED WORK

The freight logistics industry has become increasingly data-driven as organizations seek to improve operational efficiency, reduce transportation costs, and maximize profitability. Profitability in logistics depends on multiple interconnected factors, including shipment characteristics, transportation mode, operational efficiency, customer requirements, and resource utilization. Christopher [1] emphasizes that modern logistics organizations gain competitive advantages by integrating analytical decision-support systems into operational processes rather than relying solely on historical experience. Likewise, Chopra [2] explains that supply chain profitability is achieved by optimizing transportation, inventory, facilities, and operational planning while maintaining high customer service levels. Together, these studies establish the theoretical foundation for applying predictive analytics to improve financial performance in logistics operations.

The increasing availability of operational data has accelerated the adoption of predictive analytics across logistics and supply chain management. Waller and Fawcett [3] argue that data science has fundamentally transformed supply chain management by enabling organizations to convert large volumes of operational data into actionable business intelligence that supports strategic and operational decision-making. Similarly, Choi, Wallace, and Wang [4] highlight that big data analytics enhances operational visibility, forecasting accuracy, and organizational performance by extracting meaningful insights from complex logistics data. Ivanov and Dolgui [5] further demonstrate that predictive analytics and digital supply chain technologies improve operational resilience by enabling proactive and data-driven decisions under uncertain business conditions. Collectively, these studies demonstrate that predictive analytics has become an essential component of modern freight logistics.

Machine learning has emerged as one of the most effective approaches for solving complex logistics problems because shipment performance is often influenced by nonlinear interactions among operational, temporal, and shipment-related variables. Carbonneau, Laframboise, and Vahidov [6] reviewed numerous machine learning applications in supply chain management and concluded that machine learning algorithms frequently outperform traditional statistical methods when modeling complex operational systems. Although previous studies have primarily focused on demand forecasting and inventory management, their findings support the broader application of machine learning techniques to other logistics decision-making problems, including shipment profitability prediction.

Among supervised learning techniques, ensemble tree-based algorithms have consistently demonstrated superior performance for structured business datasets. Breiman [7] introduced Random Forest as an ensemble learning algorithm that improves predictive accuracy while reducing overfitting by combining

multiple decision trees. In addition to its predictive capability, Random Forest provides feature importance measures that allow researchers to identify the operational variables contributing most significantly to model predictions. Chen and Guestrin [8] later introduced XGBoost, an optimized gradient boosting framework designed to improve computational efficiency and predictive accuracy for large-scale machine learning applications. More recently, Grinsztajn, Oyallon, and Varoquaux [9] demonstrated through extensive empirical evaluation that tree-based ensemble methods continue to outperform deep learning models on many structured tabular datasets commonly encountered in industrial and business applications. These findings provide strong theoretical support for selecting Random Forest and XGBoost as the primary predictive models evaluated in this study.

Another important challenge in real-world logistics classification problems is class imbalance, where profitable shipments substantially outnumber non-profitable shipments. Learning algorithms trained on imbalanced datasets often become biased toward the majority class, reducing their ability to correctly identify minority observations that may represent financially significant business cases. To address this issue, Chawla et al. [10] proposed the Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic minority-class observations during model training to improve classifier performance without modifying the original testing data. SMOTE has become one of the most widely adopted approaches for handling imbalanced classification problems and was incorporated into this study to improve the identification of non-profitable shipments while maintaining an unbiased evaluation of model performance.

Although previous research has demonstrated the value of predictive analytics and machine learning for logistics optimization, most studies have concentrated on demand forecasting, inventory optimization, transportation planning, or supply chain resilience. Comparatively fewer studies have investigated shipment profitability prediction using operational characteristics that are available before shipment execution. This study addresses that gap by developing a practical machine learning framework that predicts shipment profitability using historical freight logistics data. By comparing Logistic Regression, Random Forest, and XGBoost, incorporating SMOTE to address class imbalance, and analyzing feature importance to identify the operational drivers of profitability, this research contributes both to the academic literature on logistics analytics and to the practical implementation of predictive decision-support systems within freight logistics organizations.

IV. METHODOLOGY

This study followed an observational quantitative research design based on historical shipment records obtained from real-world freight logistics operations. The objective was to develop and evaluate machine learning models capable of predicting shipment profitability before shipment completion using operational information available during the shipment planning process. The proposed methodology followed a complete data science workflow consisting of data collection, data preprocessing, exploratory data analysis, feature engineering, model development, model evaluation, and business interpretation.

A. Data Collection

The dataset was obtained from Gallagher Transport International, a customs brokerage and freight logistics company that manages international shipments through air, ocean, and ground transportation. Operational data were extracted from the CargoWise enterprise resource planning (ERP) system and included shipment declarations processed between December 2025 and May 2026.

The original dataset contained 8,270 shipment records and 49 variables describing transportation characteristics, shipment dimensions, operational timelines, financial information, and business attributes. After removing canceled shipments, incomplete records, and variables that introduced information leakage, the final analytical dataset consisted of 7,653 observations and 41 variables.

B. Data Preparation

Data preprocessing was performed to improve data quality and ensure consistency before model development. Missing values were handled according to their characteristics, inconsistent records were removed, and categorical variables were standardized. Date fields were converted into Python datetime objects to enable temporal feature extraction, while numerical variables were transformed into appropriate formats for analysis.

Variables containing information unavailable before shipment completion, such as financial outcome variables directly related to profitability, were excluded from model training to prevent target leakage. This ensured that the predictive models relied only on operational information available before shipment execution.

C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand shipment characteristics, identify operational patterns, and evaluate relationships among variables. Descriptive statistics and visualization techniques were used to analyze shipment volume over time, transportation mode distribution, branch activity, profitability trends, and relationships among numerical variables.

Correlation analysis was also performed to identify redundant variables and support feature selection prior to model development. These analyses provided valuable insights into shipment behavior while guiding subsequent feature engineering decisions.

D. Feature Engineering

Several new variables were engineered to improve predictive performance by capturing temporal and operational characteristics of each shipment. Temporal variables included month, quarter, and weekday, allowing the models to capture seasonal business patterns.

Operational variables included transit time, total process time, and the weight-to-volume ratio, which describe shipment efficiency and cargo characteristics. Additional shipment-related variables, including invoice counts, shipment weight, volume, packs, container count, and transportation mode, were retained because they represent information available before shipment execution and are expected to influence profitability.

E. Class Imbalance Handling

The target variable, *Profit_Flag*, classified shipments as profitable or non-profitable according to the recorded job profit. Because profitable shipments represented the majority class, class imbalance was addressed using the Synthetic Minority Over-sampling Technique (SMOTE). SMOTE was applied exclusively to the training dataset after the train-test split to generate synthetic observations for the minority class while preserving the original class distribution in the testing dataset. This approach prevented information leakage and provided an unbiased evaluation of model performance.

F. Model Development

Four classification models were evaluated throughout this study. A Baseline classifier was first established by predicting the majority class for all observations, providing a minimum performance benchmark. Three supervised machine learning algorithms were then implemented and compared: Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost).

Logistic Regression served as the linear baseline model, while Random Forest and XGBoost were selected because of their ability to capture nonlinear relationships commonly found in structured logistics datasets. Hyperparameter optimization for the Random Forest classifier was performed using GridSearchCV with five-fold cross-validation to identify the optimal model configuration.

G. Model Evaluation

Model performance was evaluated using an independent testing dataset that retained the original class distribution. Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) were used as evaluation metrics.

Confusion matrices were generated to examine classification performance for both profitable and non-profitable shipments. Feature importance analysis was subsequently performed using the optimized Random Forest model to identify the operational variables contributing most significantly to profitability prediction.

H. Technical Environment

The analysis was implemented using Python within a Jupyter Notebook environment. Data manipulation and preprocessing were performed using Pandas and NumPy, while visualization was conducted with Matplotlib and Seaborn. Machine learning models were developed using the Scikit-learn library, XGBoost was implemented through its Python API, and class imbalance was addressed using the Imbalanced-learn package. Hyperparameter optimization was performed using GridSearchCV with five-fold cross-validation to ensure robust model selection.

““latex

V. DATA ANALYSIS

A. Data Collection

The data used in this study were obtained from Gallagher Transport International, a customs brokerage and freight logistics company specializing in international freight transportation. Shipment information was extracted from the CargoWise Enterprise Resource Planning (ERP) system, which stores operational, transportation, and financial information generated throughout the shipment lifecycle.

The dataset includes shipment declarations processed between December 2025 and May 2026. Initially, 8,270 shipment records containing 49 variables were extracted from the operational database. These variables describe multiple aspects of shipment operations, including transportation mode, shipment dimensions, origin and destination locations, operational timestamps, invoice information, container details, and financial outcomes.

The primary objective of the data collection process was to obtain historical operational information that could be used to develop predictive models capable of estimating shipment profitability before shipment completion. To achieve this objective, only shipment records with sufficient operational information were retained for subsequent analysis, while canceled and incomplete records were removed during the data preparation stage.

The collected data provide a comprehensive representation of real-world freight logistics operations and form the foundation for the exploratory analysis, feature engineering, and machine learning models presented in the following sections.

B. Data Preparation

Data preparation was performed to improve data quality and ensure that the dataset was suitable for machine learning modeling. The preprocessing workflow included removing canceled shipments, eliminating incomplete records, correcting inconsistent values, and standardizing data types across all variables.

Date fields were converted into the appropriate datetime format to enable temporal analysis and the calculation of operational metrics. Categorical variables were standardized to ensure consistency across shipment records, while numerical variables were validated and converted to the appropriate data types for subsequent analysis.

Records containing missing or inconsistent information were carefully evaluated. Variables with limited missing values were cleaned using appropriate preprocessing techniques, whereas observations with insufficient information for reliable analysis were excluded from the final dataset. These procedures reduced noise and improved the overall quality of the analytical dataset.

To prevent data leakage, variables containing information that would only be available after shipment completion, particularly those directly related to the financial outcome of each shipment, were excluded from model training. Consequently, the predictive models relied exclusively on operational and shipment-related information that would be available before shipment execution.

After completing the preprocessing stage, the final analytical dataset consisted of 7,653 shipment records and 41 variables, providing a reliable and consistent foundation for exploratory analysis, feature engineering, and predictive modeling.

C. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the characteristics of the shipment data, identify operational patterns, and detect relationships among variables before model development.

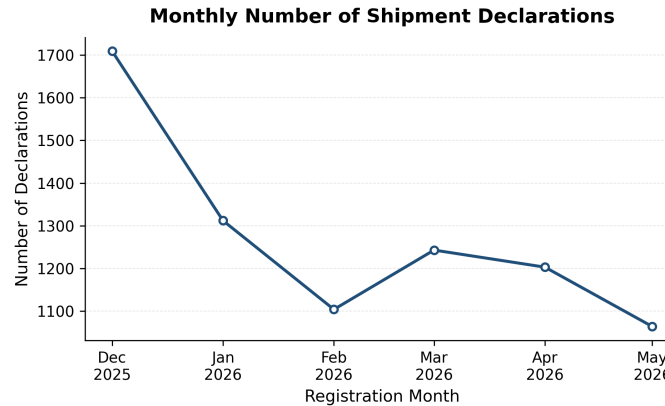


Fig. 1. Declarations processed per month

This chart shows the number of declarations over time. We observe a peak in December, followed by a significant decline in early months, suggesting seasonal demand patterns. Although there is a partial recovery in March, the overall trend remains lower compared to the year-end peak.

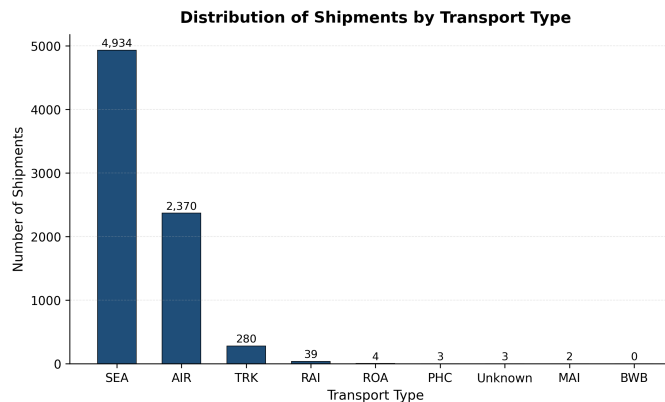


Fig. 2. Transport Type Distribution

The transport type distribution shows a strong dominance of sea shipments, followed by air transport. Other modes contribute minimally to overall operations. This indicates a high dependency on specific transport types, which may introduce operational risks, particularly in modes prone to delays such as sea transport.

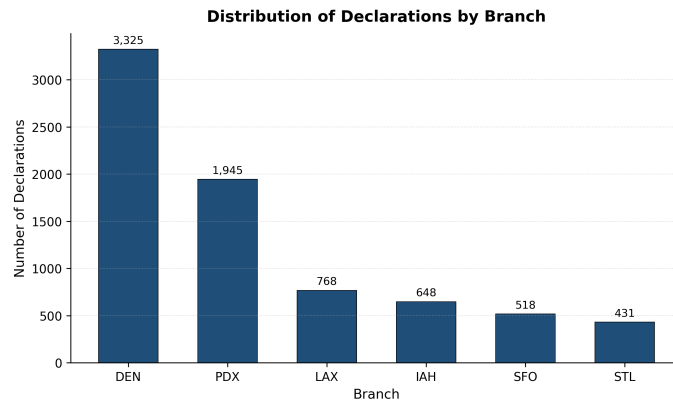


Fig. 3. Branch Distribution

This chart shows the distribution of declarations across branches. The Denver branch dominates operations, handling a significantly higher volume than all other locations. Portland represents a secondary hub, while the remaining branches contribute at a much smaller scale. This concentration may introduce operational risk and makes overall performance highly dependent on a single branch.

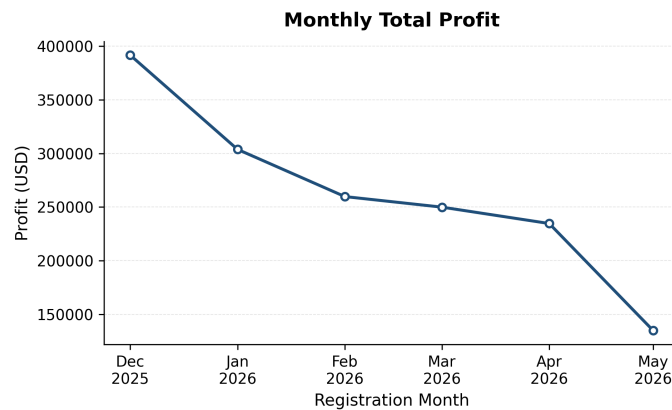


Fig. 4. Monthly Profit Trend

The monthly profit trend shows a clear decline after December, which represents the peak in profitability. This pattern closely follows the decrease in operational activity, indicating that profit is strongly linked to shipment volume. However, the sharp drop observed in later months also suggests that operational efficiency and cost factors may be affecting overall profitability.

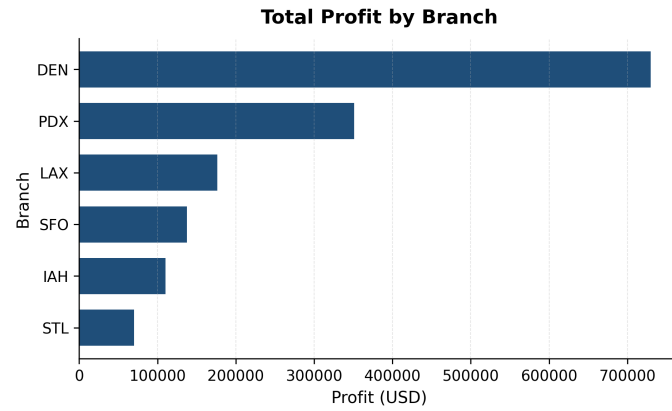


Fig. 5. Profit by Branch

The profit distribution by branch shows that Denver generates the highest profit by a large margin, followed by Portland. This aligns with the earlier observation that Denver also handles the highest operational volume, indicating a strong relationship between activity and profitability. However, this concentration suggests a dependency on a single branch, which may introduce operational and financial risk.

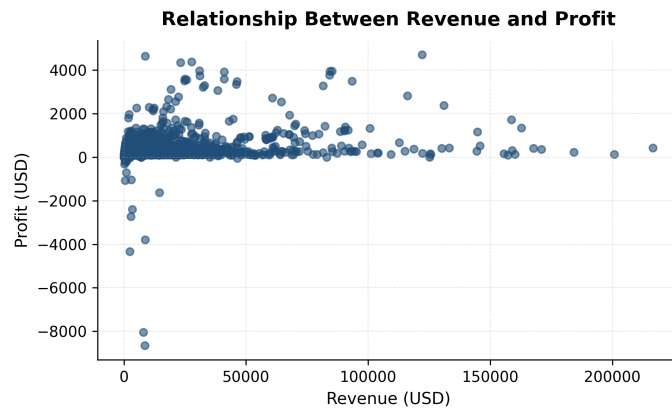


Fig. 6. Revenue vs Profit

The scatter plot shows the relationship between revenue and profit. While there is a general positive trend, the relationship is relatively weak. Notably, several cases exhibit high revenue but low or even negative profit, indicating that revenue alone does not guarantee profitability. This suggests that cost management and operational efficiency play a critical role in financial performance.

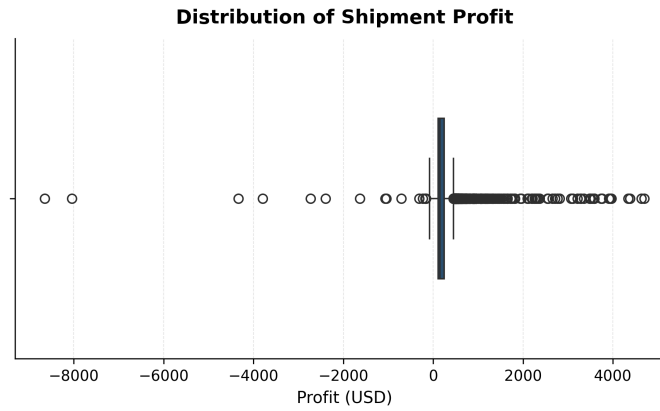


Fig. 7. Distribution of Shipment Profit

Most shipments generated positive profits, as indicated by the median value above zero. However, the distribution exhibits substantial variability and numerous outliers on both the positive and negative sides. While several shipments achieved exceptionally high profits, others experienced significant financial losses, reflecting the heterogeneous nature of freight logistics operations. Given this variability and the presence of extreme values, shipment profitability was transformed into a binary target variable, allowing the prediction task to be formulated as a supervised classification problem.

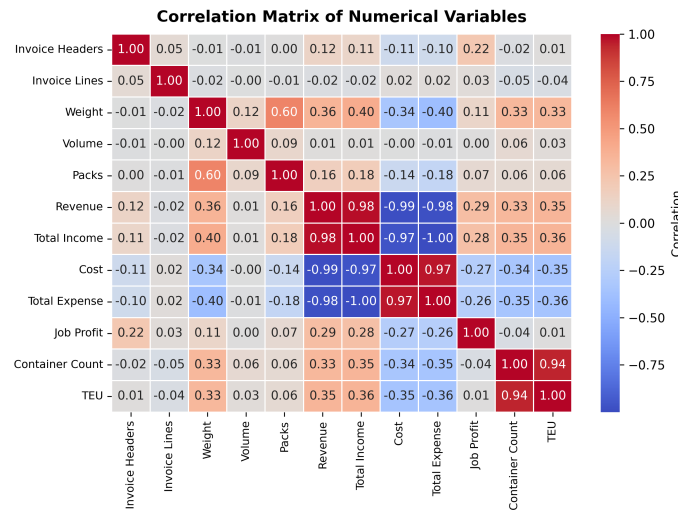


Fig. 8. Correlation Heatmap

The correlation heatmap reveals strong relationships among financial variables, particularly between revenue and total income (around 0.98), as well as strong inverse correlations with cost and total expense (around -0.97 to -0.99).

- This indicates a high level of redundancy among financial variables, as they capture closely related aspects of the same financial structure.

In contrast, job profit shows only moderate correlations with these variables (around 0.25–0.30), suggesting that profitability is not directly determined by a single financial metric.

- This highlights that high revenue does not necessarily result in high profit, reinforcing the importance of efficiency rather than scale.

Additionally, shipment size variables such as weight, packs and container count show moderate correlations with revenue (around 0.30–0.60), indicating that larger shipments tend to generate higher revenue, but not proportionally higher profitability.

- This suggests that shipment size alone is not a reliable indicator of profit.

D. Feature Engineering

Category	Features Created
Temporal	Month, Quarter, Weekday
Operational	Transit Time, Total Process Time
Shipment	Weight-to-Volume Ratio

Fig. 9. Feature Engineering

Feature engineering was performed to enhance the predictive capability of the machine learning models by transforming the original shipment data into meaningful operational and temporal features. The objective was to capture characteristics that could influence shipment profitability while ensuring that all predictor variables represented information available before shipment completion.

Several temporal variables were derived from the shipment registration date, including *month*, *quarter*, and *weekday*. These variables were introduced to capture seasonal trends and recurring operational patterns that may affect shipment demand and profitability throughout the year.

Additional operational features were generated from shipment timestamps. Transit time was calculated as the time elapsed between the estimated departure date and the actual arrival date, while total process time measured the duration between job creation and shipment completion. These variables provide indicators of operational efficiency and shipment execution performance.

A weight-to-volume ratio was also computed to better represent shipment characteristics, since cargo density often influences transportation costs and resource utilization. Original operational variables such as shipment weight, shipment volume, number of packs, invoice headers, invoice lines, container count, transportation mode, origin, destination, and branch were retained because they describe shipment complexity and operational requirements that are available before shipment execution.

To prevent information leakage, variables directly associated with the final financial outcome of each shipment, including revenue, total expense, and job profit, were excluded from the predictive features. As a result, the final feature set relied exclusively on operational information that would be available when making pre-shipment business decisions.

E. Class Imbalance Handling

The target variable, *Profit_Flag*, exhibited a noticeable class imbalance, with profitable shipments representing approximately 88% of the observations and non-profitable shipments accounting for the remaining 12%. Such imbalance can bias classification algorithms toward the majority class, resulting in poor detection of minority-class observations and misleading performance metrics.

To mitigate this issue, the Synthetic Minority Over-sampling Technique (SMOTE) [10] was applied to the training dataset after the train-test split. SMOTE generates synthetic examples of the minority class by interpolating between existing observations rather than simply duplicating them. This approach improves the model's ability to learn decision boundaries for underrepresented cases while reducing the risk of overfitting.

Applying SMOTE exclusively to the training dataset ensured that the testing dataset retained its original class distribution, providing an unbiased evaluation of model performance under realistic operating conditions. After oversampling, both classes contained an equal number of observations in the training data, allowing the machine learning models to learn representative patterns from profitable and non-profitable shipments alike.

Class Distribution Before and After SMOTE		
Training Dataset	Not Profitable (0)	Profitable (1)
Before SMOTE	737	5,371
After SMOTE	5,371	5,371

Fig. 10. SMOTE

This table illustrates the class distribution before and after applying SMOTE, demonstrating the effectiveness of the balancing process prior to model development.

F. Model Development

Following data preprocessing, feature engineering, and class balancing, several supervised machine learning algorithms were developed to predict shipment profitability. The objective was to compare models with different learning mechanisms and identify the approach that provided the best predictive performance for structured freight logistics data.

A Baseline classifier was first established by predicting the majority class for every observation. This simple model served as a performance benchmark, allowing the effectiveness of the machine learning algorithms to be evaluated against a naïve classification strategy.

Logistic Regression was implemented as the baseline linear model because of its simplicity and interpretability. As a generalized linear classifier, it estimates the probability of a shipment being profitable based on a linear combination of the predictor variables. Although widely used in binary classification problems, Logistic Regression may have limited ability to capture the complex nonlinear relationships that often characterize logistics operations.

Random Forest was selected as the primary ensemble learning algorithm due to its robustness, ability to model nonlinear interactions, and resistance to overfitting. By aggregating predictions from multiple decision trees, Random Forest can effectively capture complex operational patterns while providing feature importance measures that support model interpretability.

Extreme Gradient Boosting (XGBoost) was also evaluated as a high-performance ensemble algorithm. XGBoost builds decision trees sequentially, allowing each new tree to correct the errors of the previous ones while incorporating regularization techniques that improve generalization and computational efficiency. Its strong performance on structured tabular datasets makes it particularly suitable for logistics prediction tasks.

To improve model performance, the Random Forest classifier was further optimized using GridSearchCV with five-fold cross-validation. Hyperparameter tuning was performed to identify the optimal combination of model parameters while minimizing the risk of overfitting and improving generalization to unseen shipment data.

G. Model Evaluation

The predictive performance of the developed models was evaluated using an independent testing dataset that preserved the original class distribution. To provide a comprehensive assessment, multiple evaluation metrics were considered, including Accuracy, Precision, Recall, F1-score, and the Area Under the Receiver Operating Characteristic Curve (ROC-AUC). While accuracy provides an overall measure of classification

performance, Precision and Recall offer additional insight into the model's ability to correctly identify profitable and non-profitable shipments. The F1-score balances Precision and Recall, whereas ROC-AUC evaluates the model's capability to discriminate between the two classes across different classification thresholds.

The following table summarizes the performance of the Baseline classifier, Logistic Regression, Random Forest, and XGBoost. As expected, the Baseline model achieved a relatively high accuracy by predicting the majority class but failed to identify non-profitable shipments, demonstrating that accuracy alone is insufficient for evaluating imbalanced classification problems.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Baseline	0.88	0.88	1.00	0.94	0.50
Logistic Regression	0.75	0.92	0.78	0.85	0.70
Random Forest	0.92	0.94	0.97	0.95	0.91
XGBoost	0.89	0.96	0.92	0.94	0.90

Fig. 11. Model Comparison

Among the machine learning models, Logistic Regression produced the lowest overall predictive performance, suggesting that the relationship between operational variables and shipment profitability is not adequately represented by a linear decision boundary. In contrast, the ensemble learning methods demonstrated superior predictive capability by capturing the nonlinear interactions among shipment characteristics, operational variables, and temporal features.

Random Forest achieved the best overall balance between predictive accuracy and class discrimination after hyperparameter optimization using GridSearchCV. Its strong performance, combined with its ability to provide interpretable feature importance measures, led to its selection as the final predictive model for this study. Although XGBoost also produced competitive results, Random Forest provided slightly better overall performance while maintaining greater model interpretability, making it more suitable for supporting operational decision-making in freight logistics.

H. Feature Importance Analysis

One advantage of the Random Forest algorithm is its ability to estimate the relative importance of each predictor variable, providing valuable insight into the operational factors that influence shipment profitability. Feature importance analysis was performed using the optimized Random Forest model to identify the variables that contributed most significantly to the classification process.

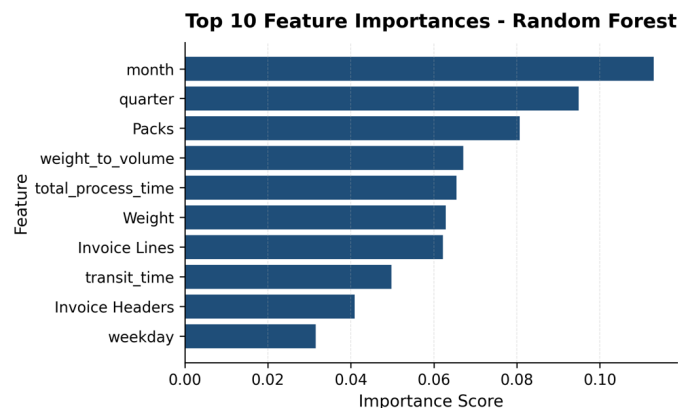


Fig. 12. Feature importance

The visualization presents the ten most important features ranked according to their contribution to the model. Temporal variables, particularly *month* and *quarter*, were identified as the strongest predictors of shipment profitability, indicating that seasonal patterns play a substantial role in freight operations. These findings suggest that fluctuations in shipment demand and operational workload throughout the year have a measurable impact on financial performance.

Operational characteristics also demonstrated considerable predictive power. Variables such as the number of packs, weight-to-volume ratio, shipment weight, and invoice complexity (represented by invoice headers and invoice lines) ranked among the most influential predictors. These variables describe shipment size, cargo composition, and documentation complexity, all of which directly affect transportation costs, handling requirements, and operational efficiency.

Process-related variables, including total process time and transit time, also contributed to model performance, indicating that operational efficiency influences shipment profitability. Longer processing or transportation times may increase operational costs and reduce overall financial performance.

Overall, the feature importance analysis demonstrates that shipment profitability is not determined by a single operational factor but rather by the interaction of temporal patterns, shipment characteristics, documentation complexity, and process efficiency. These insights enhance the interpretability of the predictive model and provide practical guidance for logistics managers seeking to improve operational planning and profitability.

I. Business Insights

The predictive framework developed in this study demonstrates that machine learning can support operational and financial decision-making in freight logistics by identifying shipments with a higher probability of being profitable before shipment execution. Rather than relying solely on historical financial reports, logistics managers can use predictive analytics to evaluate shipment characteristics during the planning stage and make proactive operational decisions.

The feature importance analysis indicates that shipment profitability is primarily influenced by seasonal patterns, shipment characteristics, and operational efficiency. Temporal variables such as month and quarter suggest that profitability varies throughout the year, highlighting the importance of anticipating seasonal demand when planning transportation capacity and resource allocation. Similarly, shipment characteristics, including cargo size, weight-to-volume ratio, and documentation complexity, provide valuable indicators of the operational effort required to process each shipment.

From a business perspective, these findings enable organizations to identify shipments that may require additional operational attention or cost control measures before execution. Predictive profitability estimates can support pricing strategies, shipment planning, workforce allocation, and operational scheduling while reducing dependence on retrospective financial analysis.

The results also demonstrate the value of using interpretable machine learning models in logistics operations. The Random Forest model not only achieved strong predictive performance but also provided transparent insights into the operational factors influencing profitability. This interpretability increases confidence in model predictions and facilitates their integration into business decision-support systems.

Overall, the proposed framework illustrates how historical operational data can be transformed into actionable business intelligence, enabling freight logistics companies to improve operational efficiency, optimize resource utilization, and support more informed and data-driven decision-making.

VI. EXPECTED OUTCOMES

This practicum project successfully developed a machine learning framework capable of predicting shipment profitability using historical operational data from freight logistics operations. The final workflow integrated multiple stages of the data science lifecycle, including data collection, data cleaning, exploratory

data analysis (EDA), feature engineering, model development, hyperparameter optimization, and business interpretation.

The data preparation process resulted in a cleaned and structured dataset containing 7,653 shipment records and 41 operational variables extracted from the CargoWise system. New temporal and operational features, including month, quarter, weekday, transit time, total process time, and weight-to-volume ratio, were engineered to improve predictive performance. Because profitable shipments largely outnumbered non-profitable shipments, the Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training dataset to address class imbalance while preserving an unbiased evaluation on the original test data.

Four classification approaches were evaluated: a Baseline classifier, Logistic Regression, Random Forest, and XGBoost. Model performance was assessed using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Hyperparameter optimization was performed using GridSearchCV with five-fold cross-validation, resulting in an optimized Random Forest classifier that achieved the best overall predictive performance.

Feature importance analysis showed that seasonality, represented by the month and quarter variables, was the strongest predictor of shipment profitability. Shipment characteristics, including weight-to-volume ratio, shipment weight, and the number of packs, together with invoice complexity and operational processing time, also contributed significantly to model predictions. These findings indicate that shipment profitability depends on the interaction of multiple operational and temporal factors rather than on a single business variable.

From a practical perspective, the proposed framework demonstrates how historical shipment data can be transformed into a decision-support tool capable of estimating shipment profitability before shipment execution. Such predictions can assist logistics managers in resource allocation, shipment planning, and operational decision-making while reducing dependence on retrospective financial analysis.

Although the proposed framework achieved strong predictive performance, several limitations remain. The study was based on shipment records from a single logistics company and did not incorporate external factors such as market conditions, fuel prices, customer behavior, or macroeconomic variables that may also influence profitability. Future research could evaluate additional machine learning algorithms, incorporate external operational data, or extend the framework to simultaneously predict shipment delays and profitability within a multi-target predictive model.

VII. TIMELINE

- 1) Week 1: Defined the project scope, business problem, research objectives, and collected shipment data from the CargoWise system.
- 2) Week 2: Performed data cleaning, handled missing values, standardized data types, removed canceled and incomplete records, and prepared the final analytical dataset.
- 3) Week 3: Conducted exploratory data analysis (EDA) to examine shipment volume, profitability trends, transportation modes, branch performance, and operational characteristics.
- 4) Week 4: Engineered temporal and operational features, including month, quarter, weekday, transit time, total process time, and weight-to-volume ratio, and prepared the modeling dataset.
- 5) Week 5: Addressed class imbalance using SMOTE, established the Baseline classifier, and prepared the training and testing datasets.
- 6) Week 6: Developed and evaluated Logistic Regression, Random Forest, and XGBoost models using Accuracy, Precision, Recall, F1-score, and ROC-AUC as evaluation metrics.
- 7) Week 7: Performed hyperparameter optimization using GridSearchCV, selected the best-performing Random Forest model, and analyzed feature importance and confusion matrix results.
- 8) Week 8: Interpreted the results, generated business insights, prepared visualizations, and completed the final practicum report and presentation.

REFERENCES

- [1] M. Christopher, *Logistics & Supply Chain Management*, 5th ed. Pearson, 2016.

- [2] S. Chopra, *Supply Chain Management: Strategy, Planning, and Operation*, 7th ed. Pearson, 2019.
- [3] M. A. Waller and S. E. Fawcett, "Data science, predictive analytics, and big data: A revolution that will transform supply chain design and management," *Journal of Business Logistics*, vol. 34, no. 2, pp. 77–84, 2013.
- [4] T.-M. Choi, S. Wallace, and Y. Wang, "Big data analytics in operations management," *Production and Operations Management*, vol. 27, no. 10, pp. 1868–1883, 2018.
- [5] D. Ivanov and A. Dolgui, "A digital supply chain twin for managing the disruption risks and resilience in the era of industry 4.0," *Production Planning & Control*, vol. 32, no. 9, pp. 775–788, 2021.
- [6] R. Carbonneau, K. Laframboise, and R. Vahidov, "Application of machine learning techniques for supply chain demand forecasting," *European Journal of Operational Research*, vol. 184, no. 3, pp. 1140–1154, 2008.
- [7] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [8] T. Chen and C. Guestrin, "Xgboost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [9] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on typical tabular data?" *Advances in Neural Information Processing Systems*, vol. 35, pp. 507–520, 2022.
- [10] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "Smote: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.